

Computer Science & Information Technology

Discrete Mathematics

DPP: 5

Set Theory and Algebra

Q1 Let $f : A \rightarrow B$ be a function, and let E and F be subsets of A . Consider the following statements.

$$S1: f(E \cup F) = f(E) \cup f(F)$$

$$S2: f(E \cap F) \subseteq f(E) \cap f(F)$$

$S3: f(E \cap F) = f(E) \cap f(F)$ if function f is one-one.

Which of the following is true?

(A) Only $S2$ and $S3$ are true.

(B) Only $S1$ and $S3$ are true.

(C) Only $S1$ and $S2$ are true.

(D) All $S1$, $S2$ and $S3$ are true.

Q2 Let $f(x) = \frac{x}{x+1}$ and $g(x) = \frac{x}{1-x}$
then $(f \circ g)^{-1}(x) = ?$

(A) x

(B) $x-1$

(C) $x+1$

(D) $x+2$

Q3 Let $A = \mathbb{R} - \{3\}$
 $B = \mathbb{R} - \{1\}$

Where $\mathbb{R}$ is the set of all real numbers if
 $f: A \rightarrow B$ such that $f(x) = \frac{x-2}{x-3}$, then

(A) f is one-one but not onto

(B) f is not one-one but onto

(C) f is bijection

(D) f is neither one-one nor onto

Q4 If the function $f: [1, \infty) \rightarrow [1, \infty)$ defined by
 $f(x) = 2^{x(x-1)}$ is invertible, then $f^{-1}(x)$ is :

(A) $\frac{1 + \sqrt{1 + 2 \log_2 x}}{4}$

(B) $\frac{1 - \sqrt{1 + 2 \log_2 x}}{4}$

(C) $\frac{1 + \sqrt{1 + 4 \log_2 x}}{2}$

(D) $\frac{1 - \sqrt{1 + 4 \log_2 x}}{2}$

Q5 The domain of the function:

$$f(x) = \sin \left\{ \log \left(\frac{\sqrt{4-x^2}}{(1-x)} \right) \right\} \text{ is :}$$

(A) $(-2, 0)$

(B) $(-2, 1)$

(C) $(-2, 2) - \{1\}$

(D) $(0, 2) - \{1\}$

Q6 Which of the following statements is /are True?

(A) A constant function is one-one iff domain of the functions has exactly one element

(B) A constant function is onto iff co-domain of the functions has exactly one element

(C) Every one-one function on a finite set A is bijection

(D) Every onto function on a finite set A is bijection

Q7 Let $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$

Set C is defined as $C = \{f: A \rightarrow B \mid 2 \in f(x) \text{ for some } x \in A, \text{ and } f \text{ is not one-one}\}$

Then number of elements in set C are _____.

Q8 Let $|A| = 6$ and $|B| = 3$, then number of onto functions possible from set A to set B are _____.

Answer Key


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Q1 (D)
Q2 (A)
Q3 (C)
Q4 (C)

Q5 (B)
Q6 (A, B, C, D)
Q7 19~19
Q8 540~540

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Hints & Solutions

Q1 Text Solution:

Statement S1: $f(E \cup F) = f(E) \cup f(F)$

To prove this, let's consider an arbitrary element $y \in f(E \cup F)$.

1. By definition, if $y \in f(E \cup F)$, there exists an element $x \in E \cup F$ such that $f(x) = y$.
2. Since $x \in E \cup F$, x must be in E or F .

If $x \in E$, then $y = f(x) \in f(E)$.

If $x \in F$, then $y = f(x) \in f(F)$.

3. In either case, $y \in f(E) \cup f(F)$.

This shows that $f(E \cup F) \subseteq f(E) \cup f(F)$.

Conversely, suppose $y \in f(E) \cup f(F)$.

1. Then $y \in f(E)$ or $y \in f(F)$.
2. If $y \in f(E)$, there exists some $x \in E$ such that $f(x) = y$.
3. Similarly, if $y \in f(F)$, there exists some $x \in F$ such that $f(x) = y$.
4. In either case, $x \in E \cup F$, so $y \in f(E \cup F)$.

Thus, $f(E \cup F) = f(E) \cup f(F)$, proving S1 is true.

Statement S2: $f(E \cap F) \subseteq f(E) \cap f(F)$

To prove this, let's consider an arbitrary element $y \in f(E \cap F)$.

1. By definition, if $y \in f(E \cap F)$, there exists an element $x \in E \cap F$ such that $f(x) = y$.
2. Since $x \in E \cap F$, $x \in E$ and $x \in F$.
3. Therefore, $f(x) \in f(E)$ and $f(x) \in f(F)$, so $y \in f(E) \cap f(F)$.

This shows that $f(E \cap F) \subseteq f(E) \cap f(F)$, proving S2 is true.

Statement S3: $f(E \cap F) = f(E) \cap f(F)$ if f is one-one

To prove this, let's assume f is injective (one-to-one) and show both directions of the equality.

(1) Proving $f(E \cap F) \subseteq f(E) \cap f(F)$:

This part is the same as in S2 and has already been proven above.

(2) Proving $f(E) \cap f(F) \subseteq f(E \cap F)$:

Let $y \in f(E) \cap f(F)$.

1. Since $y \in f(E) \cap f(F)$, there exist elements $x_1 \in E$ and $x_2 \in F$ such that $f(x_1) = y$ and $f(x_2) = y$.
2. Since f is injective, $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
3. Therefore, $x_1 = x_2 \in E \cap F$, and $y = f(x_1) = f(x_2) \in f(E \cap F)$.

Thus, $f(E) \cap f(F) \subseteq f(E \cap F)$.

Combining both directions, we get $f(E \cap F) = f(E) \cap f(F)$ if f is injective, proving S3 is true if f is one-one.

Q2 Text Solution:

$$f\left(\frac{x}{x+1}\right) = \frac{x}{x+1}, \quad g\left(\frac{x}{1-x}\right) = \frac{x}{1-x}$$

$$f \circ g\left(\frac{x}{1-x}\right) = f\left(\frac{\frac{x}{1-x}}{1-\frac{x}{1-x}}\right) = \frac{\frac{x}{1-x}}{\frac{1-x-x}{1-x}} = \frac{\frac{x}{1-x}}{\frac{1-2x}{1-x}} = \frac{x}{1-2x} = x$$

Thus, $f(g(x)) = x$ which means $f \circ g$ is the identity function.

$$\text{So, } (f \circ g)^{-1}(x) = x$$

Q3 Text Solution:

Let $f(x_1) = f(x_2)$

$$\frac{x_1-2}{x_1-3} = \frac{x_2-2}{x_2-3}$$

$$x_1 x_2 - 3x_1 - 2x_2 + 6 = x_1 x_2 - 2x_1 - 3x_2 + 6$$

$$x_1 = x_2$$

Hence, f is one - one.

$$f(x) : y = \frac{x-2}{x-3}$$

$$xy - 3y = x - 2$$

$$xy - x = 3y - 2$$

$$x = \frac{3y-2}{y-1}$$

$$x \in \mathbb{R} - \{1\}$$

Since $y \in B = \mathbb{R} - \{1\}$, x is well-defined for all $y \in B$. Thus, f is onto.

Since f is both one-one and onto, it is a bijection.

Q4 Text Solution:

$$y = 2^{x(x-1)}$$

$$\log_2 y = x(x-1)$$

$$x^2 - x - \log_2 y = 0$$

$$x = 1 \pm \frac{\sqrt{1+4\log_2 y}}{2}$$



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In terms of x it can be written as $\frac{1 \pm \sqrt{1+4\log_2 x}}{2}$
 As the range is $\left[1, \infty\right)$, $\frac{1 - \sqrt{1+4\log_2 x}}{2}$ will be
 invalid.

Hence, option c is correct.

Q5 Text Solution:

$$f(x) = \sin \left\{ \log \left(\frac{\sqrt{4-x^2}}{(1-x)} \right) \right\}$$

Due to log, $\frac{\sqrt{4-x^2}}{(1-x)} > 0$ must hold true.

Also because of square root,

$$4 - x^2 > 0 \text{ must hold true}$$

$$= (2+x)(2-x) > 0$$

$$= -2 < x < 2$$

As numerator is positive, hence the denominator must also be positive.

$$\text{Thus, } 1-x > 0$$

$$\Rightarrow x < 1$$

Hence, domain of the function is $(-2, 1)$.

Q6 Text Solution:

(A) A constant function maps every element in the domain to the same single element in the co-domain. This means it cannot be one-one if the domain has more

than one element, because multiple elements in the domain would map to the same element in the co-domain.

(B) A constant function maps every element in the domain to a single element in the co-domain. For the function to be onto, the co-domain must contain only that

single element; otherwise, it cannot cover all elements in the co-domain.

(C) A one-one function (injective function) on a

finite set A maps each element of A uniquely to the co-domain. If the function is also a bijection, it must be both

one-one and onto, meaning the co-domain must be exactly the range of A .

(D) An onto function (surjective function) on a finite set A covers the entire co-domain. Since the function is onto and A is finite, it must also be one-one (injective)

because otherwise, there would be elements in the co-domain not covered

Q7 Text Solution:

$$\text{Total number of functions from } A \rightarrow B = 4^3 = 64$$

$$\text{Number of injection functions from } A \rightarrow B = 4 \times 3 \times 2 = 24$$

$$\text{Thus, the number of non-injective function} = 64 - 24 = 40$$

If 2 is not in the range, each element in A can map only to $\{1, 3, 4\}$.

The number of non-one-one functions mapping A to $\{1, 3, 4\}$:

- Total functions from A to $\{1, 3, 4\}$ is $3^3 = 27$
- One-one functions from A to $\{1, 3, 4\}$ is $3 \times 2 \times 1 = 6$
- So, non-one-one functions mapping A to $\{1, 3, 4\}$ is $27 - 6 = 21$.

$$\text{Number of Non-one-one functions with } 2 \in \text{range}(f) = 40 - 21 = 19.$$

Q8 Text Solution:

$$\text{Number of onto functions} = \text{Total functions} -$$

$$\text{Exclusion of missing one} + \text{Inclusion of missing two} = 3^6 - 3 \cdot 2^6 + 3 \cdot 1^6$$

$$= 729 - 192 + 3$$

$$= 540$$



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